CS173: Discrete Structures
Problem Set 7

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1. (a) Show that $\forall X \forall Y (X \subseteq Y \Rightarrow |X| \leq |Y|)$.

Proof. Let X and Y be arbitrary sets, and suppose $X \subseteq Y$. We must show $|X| \leq |Y|$, i.e., that there exists an injection $f : X \to Y$.

Define $f : X \to Y$ by $f(x) := x$ for all $x \in X$. This is well-defined since $X \subseteq Y$ implies every element of X is an element of Y.

We verify f is injective. Let $x_1, x_2 \in X$ and assume $f(x_1) = f(x_2)$. By definition of f, this gives $x_1 = x_2$. Therefore f is injective, and so $|X| \leq |Y|$.

Since X and Y were arbitrary, $\forall X \forall Y (X \subseteq Y \Rightarrow |X| \leq |Y|)$. □

(b) Let X and Y be sets such that $X \neq \varnothing$ and $Y \neq \varnothing$, and let $f : X \to Y$ be an injection. Prove that there exists a function $g : Y \to X$ such that $g \circ f = \mathbf{id}_X$.

Proof. Since $X \neq \varnothing$, fix an arbitrary element $x_0 \in X$.

Define $g : Y \to X$ by: for each $y \in Y$,

$$g(y) := \begin{cases} x & \text{if } \exists x' \in X \text{ s.t. } f(x') = y, \\ x_0 & \text{otherwise.} \end{cases}$$

This is well-defined as, if $\exists x \in X$ s.t. $f(x) = y$, then that x is unique by injectivity of f, so $g(y)$ is determined unambiguously. Otherwise $g(y) = x_0 \in X$.

We verify $g \circ f = \mathbf{id}_X$. Let $x \in X$ be arbitrary. Then $f(x) \in Y$ and x itself witnesses $\exists x' \in X$ s.t. $f(x') = f(x)$, so $g(f(x)) = x$ by definition of g. Therefore:

$$(g \circ f)(x) = g(f(x)) = x = \mathbf{id}_X(x).$$

Since x was arbitrary, $g \circ f = \mathbf{id}_X$. □

2. (a) Prove that $\forall A \forall B \forall C\, ((|A| \leq |B| \wedge |B| \leq |C|) \Rightarrow |A| \leq |C|)$.

Proof. Let A, B, C be arbitrary sets, and suppose $|A| \leq |B|$ and $|B| \leq |C|$. By definition of $\leq$ on cardinalities, there exist injections $f : A \to B$ and $g : B \to C$.

Define $h : A \to C$ by $h := g \circ f$, i.e. $h(a) = g(f(a))$ for all $a \in A$.

We verify h is injective. Let $a_1, a_2 \in A$ and suppose $h(a_1) = h(a_2)$, i.e. $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. Therefore h is injective, so $|A| \leq |C|$. $\square$

(b) Prove that $\forall A \forall B \forall C\, ((|A| \geq |B| \wedge |B| \geq |C|) \Rightarrow |A| \geq |C|)$.

Proof. Let A, B, C be arbitrary sets, and suppose $|A| \geq |B|$ and $|B| \geq |C|$. By definition, there exist surjections $f : A \to B$ and $g : B \to C$.

Define $h : A \to C$ by $h := g \circ f$.

We verify h is surjective. Let $c \in C$ be arbitrary. Since g is surjective, there exists $b \in B$ with $g(b) = c$. Since f is surjective, there exists $a \in A$ with $f(a) = b$. Then $h(a) = g(f(a)) = g(b) = c$.

Since c was arbitrary, h is surjective, so $|A| \geq |C|$. $\square$

(c) Prove that $\forall A \forall B \forall C\, ((|A| = |B| \wedge |B| = |C|) \Rightarrow |A| = |C|)$.

Proof. Let A, B, C be arbitrary sets, and suppose $|A| = |B|$ and $|B| = |C|$. By Cantor-Schröder-Bernstein theorem, $|A| \leq |B|$, $|B| \leq |A|$, $|B| \leq |C|$, and $|C| \leq |B|$.

By part (a), $|A| \leq |B|$ and $|B| \leq |C|$ gives $|A| \leq |C|$.

By part (a) again, $|C| \leq |B|$ and $|B| \leq |A|$ gives $|C| \leq |A|$.

By Cantor-Schröder-Bernstein theorem, $|A| \leq |C|$ and $|C| \leq |A|$ gives $|A| = |C|$. $\square$

3. For any finite sets $\boldsymbol{A}$ and $\boldsymbol{B}$, prove that $|\boldsymbol{A} \cup \boldsymbol{B}| = |\boldsymbol{A}| + |\boldsymbol{B}|$ when $\boldsymbol{A} \cap \boldsymbol{B} = \varnothing$.

   Proof. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be finite sets with $\boldsymbol{A} \cap \boldsymbol{B} = \varnothing$.

   Since $\boldsymbol{A} \cap \boldsymbol{B} = \varnothing$, we have $(\boldsymbol{A} \cup \boldsymbol{B}) \setminus \boldsymbol{A} = \boldsymbol{B}$. Since $\boldsymbol{A}$ and $\boldsymbol{B}$ are finite, $\boldsymbol{A} \cup \boldsymbol{B}$ is finite. Since $\boldsymbol{A} \subseteq \boldsymbol{A} \cup \boldsymbol{B}$, by the theorem $(\boldsymbol{Y} \subseteq \boldsymbol{X} \Rightarrow |\boldsymbol{X} \setminus \boldsymbol{Y}| = |\boldsymbol{X}| - |\boldsymbol{Y}|)$ with $\boldsymbol{X} = \boldsymbol{A} \cup \boldsymbol{B}$ and $\boldsymbol{Y} = \boldsymbol{A}$:

   $$|(\boldsymbol{A} \cup \boldsymbol{B}) \setminus \boldsymbol{A}| = |\boldsymbol{A} \cup \boldsymbol{B}| - |\boldsymbol{A}|.$$

   Since $(\boldsymbol{A} \cup \boldsymbol{B}) \setminus \boldsymbol{A} = \boldsymbol{B}$, we have $|\boldsymbol{B}| = |\boldsymbol{A} \cup \boldsymbol{B}| - |\boldsymbol{A}|$, and therefore $|\boldsymbol{A} \cup \boldsymbol{B}| = |\boldsymbol{A}| + |\boldsymbol{B}|$.

4. For any finite set X, show that $|\mathbb{P}(X)| = 2^{|X|}$.

Proof. We proceed by induction on $|X| = n$.

Basis Step ($n = 0$): If $|X| = 0$ then $X = \varnothing$, so $\mathbb{P}(X) = \{\varnothing\}$, which has cardinality $1 = 2^0$. So $|\mathbb{P}(\varnothing)| = 1 = 2^0$.

Inductive Hypothesis: Suppose that for some $k \in \mathbb{N}$, every finite set Y with $|Y| = k$ satisfies $|\mathbb{P}(Y)| = 2^k$.

Inductive Step: Let X be a finite set with $|X| = k + 1$. Fix an element $x_0 \in X$, and let $X' \coloneqq X \setminus \{x_0\}$, so $|X'| = k$ by the corollary.

Partition $\mathbb{P}(X)$ into two disjoint sets:

$$\mathcal{A} \coloneqq \{S \in \mathbb{P}(X) \mid x_0 \notin S\},$$
$$\mathcal{B} \coloneqq \{S \in \mathbb{P}(X) \mid x_0 \in S\}.$$

By Problem 3, $|\mathbb{P}(X)| = |\mathcal{A}| + |\mathcal{B}|$ since $\mathcal{A}$ and $\mathcal{B}$ are disjoint $\mathcal{A} \cap \mathcal{B} = \varnothing$,.

Counting $\mathcal{A}$: A subset $S \subseteq X$ with $x_0 \notin S$ is exactly a subset of X'. Thus the map $S \mapsto S$ is a bijection $\mathcal{A} \to \mathbb{P}(X')$. By the inductive hypothesis, $|\mathcal{A}| = |\mathbb{P}(X')| = 2^k$.

Counting $\mathcal{B}$: A subset $S \subseteq X$ with $x_0 \in S$ is exactly a set of the form $T \cup \{x_0\}$ for some $T \subseteq X'$. Thus the map $T \mapsto T \cup \{x_0\}$ is a bijection $\mathbb{P}(X') \to \mathcal{B}$.

Injective: If $T_1 \cup \{x_0\} = T_2 \cup \{x_0\}$ then $T_1 = T_2$ (removing x_0 from both sides, noting $x_0 \notin T_i$ since $T_i \subseteq X'$ and $x_0 \notin X'$).

Surjective: For any $S \in \mathcal{B}$, let $T = S \setminus \{x_0\} \subseteq X'$; then $T \cup \{x_0\} = S$.

Therefore $|\mathcal{B}| = |\mathbb{P}(X')| = 2^k$.

Combining:
$$|\mathbb{P}(X)| = |\mathcal{A}| + |\mathcal{B}| = 2^k + 2^k = 2^{k+1}.$$

By induction, $|\mathbb{P}(X)| = 2^{|X|}$ for all finite sets X.